Hypothesis Testing and Decision Errors

Lectures 20–21

QBA201 – Quantitative Business Analysis

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The purpose of a hypothesis test

- Define the population parameter and the claim.
- Choose a test and significance level before examining the result.
- Compare the observed statistic with its null distribution.
- Report the decision and its limits.

This deck begins with mean tests when the population SD is known.

Rejecting and failing to reject are different conclusions

For a production process:

$$H_0 : \mu = 500, \qquad H_1 : \mu \neq 500.$$

Reject H_0 when the data provide sufficiently strong evidence against it.

Otherwise fail to reject H_0.

Failure to reject does not establish that the mean is exactly 500.

Two error types refer to the true population state

Decision	H_0 true	H_0 false
Reject	Type I error	Correct detection
Fail to reject	Correct nonrejection	Type II error

α controls Type I error under the null.

β is Type II probability at a specified alternative. Power is $1 - \beta$.

The alternative determines the direction

Test	Null	Alternative
Two-sided	$\mu = c$	$\mu \neq c$
Left-tailed	$\mu \geq c$	$\mu < c$
Right-tailed	$\mu \leq c$	$\mu > c$

For one-sided tests, calculate the reference distribution at boundary $\mu = c$.

Discuss: Why should the direction be chosen before seeing the sample mean?

Evidence is measured relative to sampling variability

For an independent normal sample with known population SD, use

$$z_{\text{obs}} = \frac{\bar{x} - c}{\sigma/\sqrt{n}}.$$

Distance from the null needs a scale. A large-sample normal approximation can also justify this procedure under suitable conditions.

A significance level calibrates the rejection rule

For a two-sided known-SD normal test:

$$\text{Reject when } |\bar{X} - c| \geq B.$$

Choose B so the rejection probability under $\mu = c$ is the desired α.

α is chosen before observing this sample's test result.

A rejection rule with a known error probability

Assume normal fill volumes with known $\sigma = 12$ ml and $n = 64$.

$$H_0 : \mu = 500, \qquad H_1 : \mu \neq 500.$$

Rule: reject when $|\bar{X} - 500| \geq 3$ ml.

Discuss: Calculate the probability of rejection when H_0 is true.

Complete the Type I calculation

$$\mathrm{SE} = 12/\sqrt{64} = 1.5, \qquad 3/\mathrm{SE} = 2.$$

$$\alpha = \mathrm{P}(|Z| \geq 2) = 2[1 - \Phi(2)] = 0.0455.$$

The prescribed rule has Type I probability about 4.55%.

The known-SD two-sided rule

$$z_{\text{obs}} = \frac{\bar{x} - c}{\sigma/\sqrt{n}}.$$

Reject if:

$$|z_{\text{obs}}| \geq z_{\alpha/2}.$$

Equivalently, reject if the corresponding $(1 - \alpha)$ confidence interval excludes c.

One-sided rules keep the specified direction

Alternative	Rejection rule	At $\alpha = .05$
$\mu < c$	$z_{\text{obs}} \leq -z_\alpha$	$z_{\text{obs}} \leq -1.6449$
$\mu > c$	$z_{\text{obs}} \geq z_\alpha$	$z_{\text{obs}} \geq 1.6449$

Discuss: Why does using $|z|$ lose information needed for a one-sided decision?

Three equivalent ways to make the decision

1. Compare $\bar{x}$ with its calibrated rejection region.
2. Compare the standardized statistic with its critical value.
3. Compare the p-value with α.

The p-value is calculated under the null. It is not the probability that the null is true.

The p-value must match the alternative

Alternative	p-value
$\mu \neq c$	$2[1 - \Phi(\lvert z_{\text{obs}} \rvert)]$
$\mu < c$	$\Phi(z_{\text{obs}})$
$\mu > c$	$1 - \Phi(z_{\text{obs}})$

Reject when $p \leq \alpha$.

Discuss: Can a large positive z-score support a left-tail alternative?

A two-sided decision at ten-percent significance

For $H_0 : \mu = 0$ versus $H_1 : \mu \neq 0$, suppose $z_{\text{obs}} = 1.7$.

$$|1.7| > 1.6449.$$

Reject at $\alpha = .10$.

$$p = 2[1 - \Phi(1.7)] = 0.0891.$$

The p-value leads to the same decision.

Keep one-tail and two-tail critical values distinct

α	One-tail z_α	Two-tail $z_{\alpha/2}$
.10	1.2816	1.6449
.05	1.6449	1.9600
.01	2.3263	2.5758

Discuss: Why does 1.6449 appear in two different places?

Constructed salary example: a two-sided question

Suppose annual salary observations have known $\sigma = 30,000$, $n = 100$, and $\bar{x} = 185,400$.

$$H_0 : \mu = 180,000, \quad H_1 : \mu \neq 180,000, \quad \alpha = .10.$$

$$\text{SE} = 3,000, \quad z = 1.8, \quad p = 0.0719.$$

Reject at 10%, but not at 5%.

Constructed salary example: a left-tail question

Use $n = 100$, known $\sigma = 30,000$, and $\bar{x} = 174,600$.

$$H_0 : \mu \geq 180,000, \qquad H_1 : \mu < 180,000.$$

$$z = -1.8, \qquad p = \Phi(-1.8) = 0.0359.$$

Reject at $\alpha = .05$. The data provide evidence that the mean is below 180,000 under this model.

A mean claim is different from a rarity claim

If the claim is that *mean* salary is at most 200,000:

$$H_0 : \mu \leq 200,000, \quad H_1 : \mu > 200,000.$$

For constructed $\bar{x} = 205,400$, $n = 100$, $\sigma = 30,000$:

$$z = 1.8, \quad p = 0.0359.$$

Reject at 5%: evidence supports a population mean above 200,000.

A claim that individuals rarely exceed 200,000 concerns a tail proportion, not simply the mean.

Similar decision logic can use different standard errors

Known-SD mean

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

One population proportion

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}.$$

The proportion test uses the null value p_0 in its standard error.

Type II probability depends on the alternative

$$\beta(\mu_1) = \mathrm{P}_{\mu=\mu_1}(\text{fail to reject } H_0).$$

Specify the rejection rule, sample size, variability and true alternative value before calculating.

Different alternative means generally give different β values.

Power is a function of the alternative, not a universal property of a test name.

A calibrated power illustration

Consider a right-tail test at boundary $\mu_0 = 170$ with known SE $= 3.25$.

$$\alpha = .05: \quad c = 170 + 1.644854(3.25) = 175.3458.$$

At true mean 180:

$$\beta = \Phi\left(\frac{175.3458 - 180}{3.25}\right) = 0.0761.$$

Power is $1 - \beta = 0.9239$.

A stricter alpha lowers power at the same alternative

Keep $\mu_0 = 170$, true mean 180 and SE $= 3.25$.

$$\alpha = .01: \quad c = 170 + 2.326348(3.25) = 177.5606.$$

$$\beta = \Phi\left(\frac{177.5606 - 180}{3.25}\right) = 0.2265.$$

Power is 0.7735.

Discuss: Why does moving the rejection cutoff upward make detection harder?

More observations can improve the tradeoff

At fixed population SD:

$$\text{SE} = \sigma/\sqrt{n}.$$

A larger valid sample reduces SE and can increase power at a fixed α for a fixed nonnull alternative.

Better sampling and measurement remain essential. More biased observations are not a substitute for a credible design.